



Republic of Rwanda
Ministry of Education



CADASTRAL SURVEYING

LSVCS502

Perform Cadastral Surveying

Competence

RQF Level: 5

Learning Hours 100



Credits: 10

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV5002-TVET Certificate V in Land Surveying

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Purpose statement	This module describes knowledge, skills and attitudes required to perform cadastral surveying. At the end of this module, learner will be able to plan the work, collect cadastral data, produce cadastral plans, and perform boundary stake out.					
Leaning assumed to be in place	Apply land law and regulations					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria	
1. Plan the work	1.1.	Basics of cadastral surveying are properly described according to the Surveying Legal framework
	1.2.	Site reconnaissance is properly conducted based on the site conditions
	1.3.	Resources are properly allocated according to project duration
2. Collect data	2.1.	Surveying method is correctly selected based on the type of work
	2.2.	Surveying instrument is properly set up according to procedures
	2.3.	Data are accurately captured according to the selected method.
3. Produce cadastral plans	3.1.	Data are clearly prepared according to data pre-processing procedures
	3.2.	Data are correctly processed according to rules and regulation requirements.
	3.3.	Cadastral plan is properly published according to the format required
	3.4.	Cadastral report is properly prepared in line with the cadastral reporting guidelines
	4.1.	Setting out data is correctly generated based on the cadastral plan

4. Perform boundary stake out	4.2. Surveying instrument is properly set up according to specifications
	4.3. Boundaries are accurately marked based on Cadastral plan.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describing work planning procedure ✓ Describing principles of cadastral survey ✓ Identifying Surveying tools, equipment and materials ✓ Describing cadastral surveying procedures ✓ Describing surveying rules and regulations ✓ Describing computations of cadastral data ✓ Identifying elements of cadastral plan ✓ Describing principles of safety, health and security at workplace	✓ Designing work plan ✓ Operating surveying instruments ✓ Using surveying software for cadastral data processing ✓ Creating plans and maps ✓ Interpreting cadastral plans and maps ✓ Applying adjustments methods of errors correction in surveying ✓ Applying techniques of staking out ✓ Producing reports ✓ Applying communication skills ✓ Applying safety precaution at workplace	✓ Being organized and attentive ✓ Having problem solving attitude ✓ Having analytical thinking ✓ Having teamwork spirit ✓ Having time management attitude ✓ Being decision maker ✓ Being responsible ✓ Being flexible ✓ Being patient ✓ Being prompt

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Plan the work 2. Collect data 3. Produce cadastral plans 4. Perform boundary stake out
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Learning outcome 1: Plan the work		Learning hours: 20 hours
Indicative content		
● Describing cadastral surveying ✓ Concept of cadastre ✚ Definition ✚ Types of cadastres ✚ Functions ✓ Concept of cadastral surveying ✚ Objectives of cadastral surveying ✚ Types of boundaries ✓ Users of cadastral data ✚ Public institutions ✚ Private institutions ● Carrying out site reconnaissance ✓ Preparing for site visit ✓ Collecting site information ✓ Producing report ● Allocating Resources ✓ Human resources ✓ Transport ✓ Budget ✓ Equipment ✓ Tools ✓ Materials		
Resources required for the indicative content		
Equipment	GNSS Receivers, Computer, PPE,	
Materials	Field book, pens, pegs, nails, warning tape, marker pen, pencil, paint, first aid kit, beacon.	
Tools	Hammer, ranging rod, tape measures, arrows, compass, communication radios, pickaxes	
Facilitation techniques	Group discussion, scenario, simulation, individual activity ● Trainer guides learners to form small groups and asks them to discuss on the basics of cadastral surveying. Trainer asks learners to present the work done, there after provides an expert view. ● Trainer provides scenario about cadastral surveying, and asks the learners to conduct site reconnaissance in manageable groups, and present the reconnaissance report of their findings. ● Trainer asks the learners in their groups to allocate resources in line with surveying activities to be done, based on the scenario given, and the site reconnaissance findings.	

Formative assessment methods	• Written assessment • Performance assessment • Product assessment
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Learning outcome 2: Collect data		Learning hours: 30 hours
Indicative content		
• Selecting surveying methods ✓ Ground survey methods ✚ CORS (receiver and CORS base station) ✚ RTK (base and rover) ✚ Total station (traversing, point positioning techniques) ✚ Static ✓ Aerial methods ✚ Photogrammetric methods • Setting the instrument ✓ Centring the instrument ✓ Checking instrument connectivity ✓ Configuring the instrument • Capturing Data ✓ Take measurement ✓ Check accuracy ✓ Record measurement		
Resources required for the indicative content		
Equipment	GNSS Receivers, Total station, Computer, PPE,	
Materials	Surveyor's vest, field book, pens, cement, crushed stones, pegs, nails, sand, water, warning tape, marker pen, pencil, paint, safety cones, flags, first aid kit,	
Tools	Hammer, ranging rod, plumb bob, tape measures, arrows, prisms, compass, reflector, tripod, communication radios, brushes, spray gun	
Facilitation techniques	Group discussion, scenario, individual activity, Field practice, presentation, video show • Trainer displays a video that explains cadastral surveying procedures and asks learners to discuss what they observed from the video and pictures. • Trainer provides a scenario highlighting cadastral surveying work and asks the learners in manageable groups to go out and collect data of the land parcel based on the produced work plan.	
Formative assessment methods	• Written assessment • Performance assessment	

Learning outcome 3: Produce cadastral plans		Learning hours: 30 hours
Indicative content		
• Preparing data ✓ Transferring data into computer ✚ Downloading data/Typing ✚ Editing ✓ Uploading data into software ✓ Cleaning data ✓ Checking accuracy • Processing data ✓ Setting reference system ✓ Applying software processing procedures ✓ Checking accuracy • Publishing cadastral plan ✓ Selecting file format ✓ Online publishing ✓ Transferring to storage device ✓ Plotting • Producing cadastral report ✓ Preparing field survey report ✓ Consulting abutting land owner(s) for signatures ✓ Attaching deed plan		
Resources required for the indicative content		
Equipment	DGPS, Total station, Computer, PPE,	
Materials	Overall, field book, pens, cement, crushed stones, pegs, nails, sand, water, warning tape, marker pen, pencil, paint, safety cones, flags, first aid kit,	
Tools	Hammer, ranging rod, plumb bob, tape measures, arrows, prisms, compass, reflector, tripod, communication radios.	
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer provides procedures on how to transfer data into the computer, clean data, processing procedures and checking of accuracy and asks them to use collected data to produce cadastral plan of land parcel. • Trainer asks the learners in small groups to discuss on types of file formats and publish the cadastral plans and present their work and trainer provides feedback and input and asks them to refine their work if necessary.	

Formative assessment methods	• Written assessment • Performance assessment • Production assessment
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Learning outcome 4: Perform boundary stake out		Learning hours: 20 hours	
Indicative content			
● Generating setting out data ✓ Referring to existing data from land information system ✓ Extracting cadastral data ✓ Entering data into the instrument ● Setting the instrument ✓ Centring the instrument ✓ Checking instrument connectivity ✓ Configuring the instrument ● Marking boundary ✓ Setting the instrument ✓ Staking out the points ✓ Marking the point (place the beacon) ✓ Checking accuracy of setting out ✓ Produce stake out report			
Resources required for the indicative content			
Equipment		DGPS, Total station, Computer, PPE,	
Materials		Overall, field book, pens, cement, crushed stones, pegs, nails, sand, water, warning tape, marker pen, pencil, paint, safety cones, flags, first aid kit,	
Tools		Hammer, ranging rod, plumb bob, tape measures, arrows, prisms, compass, reflector, tripod, communication radios.	
Facilitation techniques		Scenario, individual activity ● Trainer displays a video on boundary/point stakeout procedure using different equipment (Total station, DGPS) and asks learners to discuss on stakeout procedure ● Trainer provides a scenario of land parcels and asks students to extract boundary coordinates.	

	• Trainer asks the learners to input the extracted coordinates into the surveying instrument. • Trainer asks learners to go out and stake out and monument the boundary points and produce report.
Formative assessment methods	• Written assessment • Performance assessment • Product assessment

Integrated/Summative assessment

Integrated situation

Due to the constraint of obtaining build permit, XYZ Ltd has been requested by Mr. Kagabo, a resident of Karama Village, Rusera cell, Rubavu Sector in Rubavu District to produce a deed plan for his land parcel where he wants to construct a residential house. The plot has an area of 400m². As a surveyor assistant employed by XYZ Ltd, you have been requested to collect all required data to produce the deed plan of the area within eight (8) hours. All tools, materials and equipment are provided by XYZ Ltd.

Resources

Tools	Hammer, panga, ranging rod, plumb bob, tape measure, arrows, prisms, calculator, reflector, tripod, communication radios.
Equipment	Differential GNSS Receiver, Total station, PPE, computer
Materials/ Consumables	Overcoat, overall, field book, pens, pegs, warning tape, marker pen, pencil, safety cones,

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Plan the work (22%)	1.1. Site reconnaissance is properly conducted based on the site conditions.	1. PPE worn			5
		2. Site information is collected			10
	1.2 Resources are properly allocated according to project duration	1. Resources are allocated			7
Learning outcome 2: Collect data (44%)	2.1 Surveying method is correctly selected based on the type of work	1. Survey method is selected			6
	2.2 Surveying instrument is properly set up according to specifications	1. Equipment is set up			7
		2. Instrument is configured			8
	2.3. Data are accurately captured according to the selected method.	1. Measurements are taken			7
		2. Accuracy checks are applied			6
		3. Measurements are recorded			10
Learning outcome 3: Produce cadastral plans (34%)	3.1. Data are clearly prepared according to data pre-processing procedures	1. Data is transferred into computer			4
		2. Data is cleaned			3
		3. Accuracy of data is checked			3
		1. Reference system is set			3
		2. Software processing procedures is performed			4

	3.2. Data are correctly processed according to rules and regulation requirements.	3. Accuracy of data is checked			3
	3.3. Cadastral plan is properly published according to the format required	1. File format selected correctly			4
		2. File transferred to storage device correctly			3
		3. Printer selected properly			3
	3.4. Cadastral report is properly prepared in line with the cadastral reporting guidelines	1. Cadastral report produced properly			4
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. International Federation of Surveyors (1998) -FIG Statement on Cadastre
<https://www.fig.net/resources/publications/figpub/pub11/FIG%20Statement%20on%20the%20Cadastre.pdf>
2. Schermerhorn, W., & Witt, G. F. (1953). Photogrammetry for cadastral survey. Photogrammetria, 10, 45-57.
3. Cadastral Information Division Department of Survey and Mapping (2022). Guideline for using GNSS-RTK in Cadastral Surveying.
[Guideline for using GNSS-RTK in Cadastral Surveying v1.pdf \(nlcs.gov.bt\)](#)